

Problem Set 6

ECON 2010 – Macroeconomics

Spring 2018

Due at the start of class April 25, 2018. You have **two weeks** to complete this problem set. **This is the final problem set!** No late problem sets will be accepted. Please refer to the syllabus section on late assignments.

This problem set will work the same as the first problem set and all problem sets. Work on a separate sheet of paper on your own, and we will correct (some of) the problem set together in class.

Name:

Problem set instructions.

1. **Do not write on this problem set sheet** until we go over the problem set during class. You will use this problem set sheet to correct the answers that you provide on the separate sheet when we go over the problem set in class the day that it is due.
 2. Write your final answers to each question on a separate sheet of paper.
 3. Write your final answers in pen.
 4. Ensure that your name appears on this problem set and on the separate sheet(s) of paper with your answers, and staple your answers to the back of this problem set.
 5. When appropriate, show your work for each problem. Even if your answer is technically incorrect, you may get many points for the work you did to arrive at that answer. If in doubt, show your work.
 6. Note that not all questions have *one* right answer. Some questions might have many correct answers.
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Problem 1 (17 points)

In the following problem you'll consider the *political* economy of the two types of government policy that can effect the aggregate economy — fiscal policy and monetary policy.

- (a) Indicate whether following statement is **true** or **false**, and then in a few sentences, **briefly explain your answer**. (4 points)

The people in government responsible for making monetary policy are elected representatives subject to direct accountability by the people (the electorate).

- (b) Indicate whether following statement is **true** or **false**, and then in a few sentences, **briefly explain your answer.** (4 points)

The institutions responsible for making fiscal policy have a specific mandate to do two things: keep price levels stable and promote economic growth.

- (c) Indicate whether following statement is **true** or **false**, and then in a few sentences, **briefly explain your answer.** (4 points)

Government bonds are sold so that the government can lend money to the economy. It's ordinary people like you and me who buy the bonds who are borrowing from the government.

- (d) Indicate whether following statement is **true** or **false**, and then in a few sentences, **briefly explain your answer.** (4 points)

The Federal Reserve Bank sets the Federal Funds (Interest) Rate by declaring in a quarterly

memo, accompanied by a news conference, what that interest rate is. All banks in the federal reserve system have to abide by the interest rate declared in this quarterly memo.

- (e) Construct your own normative statement about the politics or goals of *monetary* policy makers. You are being graded on your ability to write a normative statement, not on whether I agree or disagree with your statement. *However*, if your statement includes anything factual, those facts should be correct. **(1 point)**

Problem 2 (21 points)

In this problem, you will consider the role that banks play in affecting the money supply. Consider a simple economy with two individuals — Krystal and Clint — and two different banks — First Bank of Nashville, and Memphis Credit Union (simplifying assumption!).

Note that when the problems ask you to consider the economy as it was in the previous part of the problem, but with an addition/change, you're being asked to evaluate the **cumulative**. That is take into account everything that you've been told in previous parts of the problem, with the addition of what happens in that part of the problem. Hold all else fixed (*ceteris parabis*!) when you consider each change.

- (a) At the beginning of time period 1, Krystal and Clint each get paid from their jobs. Krystal gets a check from her employer who has a checking account at First Bank of Nashville. The check she receives is for the amount of \$500. She keeps that check in her wallet for the time being. Clint receives \$250 in Cash from his employer. If this is all you know about the

economy, what is the total money supply in the economy? **(2 points)**

- (b) Consider the economy as in the previous part of the problem above, but with the following addition/change. Consider that the First Bank of Nashville also has \$10,000 in reserves held in an account with the Federal Reserve Bank. If this is all you know about the economy, what is the total money supply in the economy? **(2 points)**
- (c) Consider the economy as in the previous parts of the problem above, but with the following addition/change. Now, Krystal takes her \$500 check to her own bank — Memphis Credit Union — and deposits it in her personal checking account. She also withdraws \$100 to keep as spending money in cash in her wallet. If this is all you know about the economy, what is the total money supply in the economy? **(2 points)**

- (d) Consider the economy as in the previous parts of the problem above, but with the following addition/change. Clint now goes to the Memphis Credit Union to take out a loan of \$300. Say that the reserve requirement for all banks in the US as established by the Federal Reserve is 10%. Is it possible for Clint to obtain this loan (assume that he is a creditworthy borrower)? Given your answer to that question, and if this is all that you know about the economy, what is the total money supply in the economy? **(3 points)**
- (e) Consider the economy as in the previous parts of the problem above, but with the following addition/change. Clint buys two gift cards, each in the amount of \$100. He also uses his credit card to make a \$100 purchase with the remaining money that he borrowed. Given your answer to that question, and if this is all that you know about the economy, what is the total money supply in the economy? **(2 points)**
- (f) Consider the economy as in the previous parts of the problem above, but with the following addition/change. If the Federal Reserve changes the reserve ratio, reducing it to 0%, and if this is all that you know about the economy, what is the total money supply in the economy? **(2 points)**

- (g) Consider the economy as in the previous parts of the problem above, but with the following addition/change. Clint returns to the Memphis Credit Union for an additional loan of \$100. Is it possible for Clint to obtain this loan (assume that he is a creditworthy borrower)? Given your answer to that question, and if this is all that you know about the economy, what is the total money supply in the economy? **(3 points)**
- (h) Consider the economy as in the previous parts of the problem above, but with the following addition/change. Say that Krystal returns to Memphis Credit Union to withdraw her previous deposit in full. Is it possible for Krystal to obtain her money? If it is not possible, give **three** distinct things that Memphis Credit Union can do in theory to (try to) return Krystal her money. I say “in theory” and “try to” because I haven’t given you enough information to know if they would actually *work*. Remember, our simple economy only exists with two individuals, two banks, the Federal Reserve, and the money supply up to this point that you have calculated. So, you need to consider the options only given this economy. **(5 points)**

Problem 4 (54 points)

In this problem, you will consider the different tools that monetary policy makers have to affect the aggregate economy. You will consider both how these tools affect the Money Market, and how they affect output and price levels in our AS-AD model (aggregate supply-aggregate demand).

- (a) Consider the one of three monetary policy tools that the Federal Reserve has at its disposal:

the reserve ratio (i.e. the reserve requirement). Answer the following questions about what happens in the Money Market and what happens in the aggregate economy when the Federal reserve **increases** the reserve ratio. **(29 points total)**

(1) In your own words, briefly explain what the reserve ratio is. **(1 point)**

(2) Consider first what happens in the Money Market after this change in Fed policy. Which curve in the Money Market shifts as a result of this change? You could say *money supply*, *money demand*, *none* or *both*. **(2 points)**

(3) In which direction does this Money Market curve shift? You can say *left*, *right*, or *no curves shift*. **(2 points)**

- (4) In a few sentences, briefly explain why the curve(s) shifted in the direction that you indicated above. Or, if no curves shifted, just write “none.” **(4 points)**
- (5) Draw a generic Money Market graph with money supply curve, and the money demand curve. Label each of your curves, and label each of your axes appropriately. Mark the start equilibrium before any curve shifts with an “A.” Draw any curve shifts that you indicated would occur with a dashed line. Label the new equilibrium (if any curves shifted) with a “B.” **(3 points)**
- (6) Indicate what has happened to the equilibrium quantity of money in the economy, and the equilibrium interest rate as a result of this Fed policy change. For **each**, you can say that it went *up*, *down*, *stayed the same*, or is *unknown*. **(2 points)**

- (7) Now that you've considered what happens in the Money Market, consider what happens in the aggregate economy. In our model of AS-AD (aggregate supply-aggregate demand), which curve shifts as a result of this change in the Money Market (brought about by the change in Fed policy)? You could say *money supply*, *money demand*, *none* or *both*. **(2 points)**
- (8) In which direction does this AD-AS curve shift? You can say *left*, *right*, or *no curves shift*. **(2 points)**
- (9) In a few sentences, briefly explain why the curve(s) shifted in the direction that you indicated above. Or, if no curves shifted, just write "none." **(4 points)**

- (10) Draw a generic AD-AS graph with both short and long run aggregate supply, and the aggregate demand curve. Label each of your curves, and label each of your axes appropriately. Mark the starting *long run* equilibrium before any curve shifts with an “A.” Draw any curve shifts that you indicated would occur with a dashed line. Focus on the *short run* resultant equilibrium, and label the new equilibrium (if any curves shifted) with a “B.” **(3 points)**

- (11) Indicate what has happened to the equilibrium quantity of money in the economy, and the equilibrium interest rate as a result of this Fed policy change. For **each**, you can say that it went *up*, *down*, *stayed the same*, or is *unknown*. **(2 points)**

- (b) Consider another of the three monetary policy tools that the Federal Reserve has at its disposal: **the (overnight) discount rate**. Answer the following questions about what happens in the Money Market and what happens in the aggregate economy when the Federal reserve **decreases** the (overnight) discount rate. **(29 points total)**

- (1) In your own words, briefly explain what the (overnight) discount rate is. **(1 point)**
- (2) Consider first what happens in the Money Market after this change in Fed policy. Which curve in the Money Market shifts as a result of this change? You could say *money supply*, *money demand*, *none* or *both*. **(2 points)**
- (3) In which direction does this Money Market curve shift? You can say *left*, *right*, or *no curves shift*. **(2 points)**
- (4) In a few sentences, briefly explain why the curve(s) shifted in the direction that you indicated above. Or, if no curves shifted, just write “none.” **(4 points)**

- (5) Draw a generic Money Market graph with money supply curve, and the money demand curve. Label each of your curves, and label each of your axes appropriately. Mark the start equilibrium before any curve shifts with an “A.” Draw any curve shifts that you indicated would occur with a dashed line. Label the new equilibrium (if any curves shifted) with a “B.” **(3 points)**
- (6) Indicate what has happened to the equilibrium quantity of money in the economy, and the equilibrium interest rate as a result of this Fed policy change. For **each**, you can say that it went *up*, *down*, *stayed the same*, or is *unknown*. **(2 points)**
- (7) Now that you’ve considered what happens in the Money Market, consider what happens in the aggregate economy. In our model of AS-AD (aggregate supply-aggregate demand), which curve shifts as a result of this change in the Money Market (brought about by the change in Fed policy)? You could say *money supply*, *money demand*, *none* or *both*. **(2 points)**

(8) In which direction does this AD-AS curve shift? You can say *left*, *right*, or *no curves shift*.
(2 points)

(9) In a few sentences, briefly explain why the curve(s) shifted in the direction that you indicated above. Or, if no curves shifted, just write “none.” **(4 points)**

(10) Draw a generic AD-AS graph with both short and long run aggregate supply, and the aggregate demand curve. Label each of your curves, and label each of your axes appropriately. Mark the starting *long run* equilibrium before any curve shifts with an “A.” Draw any curve shifts that you indicated would occur with a dashed line. Focus on the *short run* resultant equilibrium, and label the new equilibrium (if any curves shifted) with a “B.” **(3 points)**

- (11) Indicate what has happened to the equilibrium quantity of money in the economy, and the equilibrium interest rate as a result of this Fed policy change. For **each**, you can say that it went *up*, *down*, *stayed the same*, or is *unknown*. **(2 points)**

Correcting Your Problem Set (8 points)

You'll receive 8 points for correcting your problem set in class — that is, marking the answers that you got correct on this sheet, and writing an explanation for why some of your answers were incorrect on others.